Does Model Complexity Pay Off? A Rolling-Origin Benchmark for Forecasting German Household Overnight Deposit Volumes

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Abstract

Liquidity management ensures that a bank has sufficient cash to fulfill its short-term obligations while minimizing the cost of holding excess liquidity, and non-maturity deposits (NMDs) are among the most important sources of this liquidity. Despite their economic importance, forecasting NMD volumes remains methodologically underdeveloped, with no established consensus on the optimal modeling approach. This study compares two naive benchmarks, a regularized linear model, gradient-boosted trees, a feedforward neural network, and a Temporal Fusion Transformer for forecasting German household overnight deposit volumes over a 100-day horizon, using a rolling-origin evaluation across four forecast origins between 2009 and 2025 that represent distinct interest-rate regimes. All models share an identical information set with a 100-day lookback window, an identical tuning budget, and lag-augmented covariates. The regularized linear model achieved the lowest mean forecast error (MAE = 9.36 Bn. €), the lowest error relative to the constant-level benchmark (0.44), and was the only model with a positive mean R^2 (0.178), while the feedforward network and gradient-boosted trees did not improve on a one-step drift benchmark. Because only four origins are available, these differences are reported descriptively rather than as significance tests. The results indicate that increased model complexity does not translate into higher forecasting accuracy for this task, with direct implications for how banks select forecasting tools for liquidity management.

Contents

1 Forecasting Models

1.1 Forecasting Framework

We define y_t as the volume of German household overnight deposits at time t, and $\mathbf{x}_t$ as the vector of our predictors and their lagged values from Equation (1). The objective is to forecast deposit volumes over a 100-day horizon ($H = 100$) using the information available during the preceding 100 days ($L = 100$). The reasoning for $H = 100$ is because it accounts for roughly one quarter of trading days, therefore relevant for the operational liquidity management's planning cycle.

At time of forecast t the information set contains the historical deposit volume and the selected covariates over the period $t - L + 1, \ldots, t$. Each model produces the complete vector of predicted changes for the following $H = 100$ days:

$$\widehat{\boldsymbol{\Delta y}}_{t+1:t+H} = f(y_{t-L+1:t}, \mathbf{x}_{t-L+1:t}), \qquad L = 100, \quad H = 100. \tag{1}$$

Setting $L = H$ follows the common convention that the context available to a multi-horizon forecaster should be at least as long as the horizon it must predict (6; 7). Also, since the deposit series is published monthly and interpolated to a daily grid (Section **??**), a window of 100 trading days spans approximately five genuine monthly observations of the target and provides the models with recent information.

A direct multi-output forecasting strategy is used for this study. Each model estimates all 100 future changes simultaneously rather than recursively one day at a time. The predicted changes are then converted back into deposit-volume levels according to

$$\widehat{y}_{t+h} = y_t + \sum_{j=1}^{h} \widehat{\Delta y}_{t+j}, \qquad h = 1, \ldots, H. \tag{2}$$

Only information available up to time t is used for model estimation and for fitting the feature scaler, which is a robust scaler estimated on the training window alone. Actual covariate values within the 100 forecast days are never used. For the TFT, covariates are held constant at their most recently observed value over the horizon, while date-based variables that are known in advance are supplied for the entire forecast horizon.

1.2 Model Specifications

Naive benchmarks. Two naive benchmarks are used. The *one-step drift* benchmark extrapolates the single most recently observed daily change at a constant rate,

$$\hat{y}_{t_0+h} = y_{t_0} + h \cdot \Delta y_{t_0}, \qquad h = 1, \ldots, 100,$$

that is, an ARIMA$(0, 1, 0)$ process whose drift is estimated from one observation rather than from the training average. And the *constant-level* benchmark, which assumes the level remains unchanged, $\hat{y}_{t_0+h} = y_{t_0}$. Both benchmarks are fully interpretable, require no covariates, and serve as the reference against which the estimated models are compared.

Linear regression (Ridge). A multi-output ridge regression maps the flattened 100-day lookback window to the 100-step forecast vector, regularized by an ℓ_2 penalty α selected via time-series cross-validation. As a linear model, Ridge yields directly inspectable coefficients. Also, since the flattened feature vector is high-dimensional and strongly collinear, it makes shrinkage adequate.

XGBoost. Gradient-boosted regression trees are trained on the same flattened input representation (che), with maximum depth, learning rate and subsample ratio selected by chronological time-series cross-validation. Boosted trees can capture non-linear covariate interactions that Ridge cannot (has, Ch. 10), while remaining moderately interpretable.

Feedforward neural network (FFN). The FFN is a multilayer perceptron with ReLU activations trained to predict the 100-step output vector directly. The FFN is tuned under the same protocol and the same budget as Ridge and XGBoost (Section 1.3), meaning its layer configuration and ℓ_2 penalty are selected by the same chronological cross-validation. Training uses the Adam optimizer with early stopping on the chronologically last 15% of the training window, with a patience of 20 epochs and a cap of 500 epochs. Since our 100-day sequences overlap, a chronological split prevents future data from leaking into the validation set.

Temporal Fusion Transformer (TFT). Unlike the sequence models above, the TFT explicitly distinguishes between known future inputs (calendar features: weekday, month-end indicator, day-of-month) and unknown future inputs (the six macro-financial covariates). The encoder length is set to $L = 100$ to match the other models, and the model is trained with early stopping on validation loss with a patience of five validation checks. Despite being the most complex model considered, its attention mechanism was explicitly designed to offer a degree of interpretability that purely black-box models lack.

Each model is fitted $S = 5$ times to form an ensemble. Ridge, FFN and XGBoost use bootstrap-resampled training sequences together with different random seeds; repeated TFT fits differ in the random seed only. Ensemble members are combined by averaging their forecasts.

1.3 Tuning Protocol

A comparison of "complex" against "simple" models is only meaningful if the models receive comparable optimisation effort. We therefore impose a single tuning budget of at most twelve grid configurations per model, searched by grid search with a three-fold chronological `TimeSeriesSplit` and mean absolute error as the selection criterion. Hyperparameters are re-tuned at every forecast origin using only data available at that origin, so that no configuration selected on a later window is transferred to an earlier one. Table 1 documents the complete search space; the TFT architecture is fixed a priori at the configuration shown, since a grid search over the transformer was not computationally feasible within the same budget. This last asymmetry works *against* our headline conclusion rather than for it and is noted as a limitation in Section **??**.

Table 1: Hyperparameter search spaces and fixed settings.

Model	Search space / fixed settings	Configurations
Ridge	$\alpha \in \{10^{-3}, \dots, 10^{3}\}$, 8 values log-spaced	8
FFN	hidden layers $\in \{(64, 32), (128, 64)\}$; ℓ_2 penalty $\in \{10^{-4}, 10^{-3}\}$; Adam, learning rate 10^{-3}; early stopping, patience 20 epochs, max. 500	4
XGBoost	max. depth $\in \{3, 4, 5\}$; learning rate $\in \{0.05, 0.1\}$; subsample $\in \{0.7, 0.9\}$; 50 trees during search, 100 at final fit	12
TFT	fixed: hidden size 32, 4 attention heads, dropout 0.1, continuous hidden size 16, Adam, learning rate 10^{-3}, batch size 64, max. 40 epochs, early stopping patience 5, MAE loss	—
Shared: $L = 100$, $H = 100$, $S = 5$ ensemble members, robust scaling fitted on training window, seed 42.		

1.4 Covariate Specifications

To evaluate the predictive value of different economic information, each non-naive model is evaluated using the nine predefined covariate sets shown in Table 2. These specifications compare economically related variable groups, individual interest-rate variables, and subsets of the complete dataset. Each set is lag-augmented according to Equation (**??**). "Code label" column reports the identifier used in the accompanying code and in the figure filenames.

Table 2: Covariate specifications.

Specification	Included variables	Code label
All	€STR, deposit rate, geopolitical risk, inflation, DAX, and ten-year German government bond yield	`alle`
Interest rates	€STR and deposit rate	`nur_zinsen`
Financial markets	DAX and ten-year government bond yield	`nur_maerkte`
Macroeconomic	Geopolitical risk and inflation	`nur_makro`
Rates and macroeconomy	€STR, deposit rate, geopolitical risk, and inflation	`zinsen_makro`
Excluding DAX	All variables except DAX	`ohne_dax`
Excluding inflation	All variables except inflation	`ohne_inflation`
€STR only	€STR	`nur_estr`
Deposit rate only	Deposit rate	`nur_einlagezins`

1.5 Rolling-Origin Evaluation

Forecasting performance is evaluated using an expanding-window rolling-origin design. At each forecast origin o_k, all observations available up to that date form the training sample, while the following $H = 100$ days form the evaluation window:

$$\mathcal{T}_k = \{1, \ldots, o_k\}, \qquad \mathcal{E}_k = \{o_k + 1, \ldots, o_k + H\}.$$

The models are re-estimated, and their hyperparameters re-selected, at every origin as newly observed data becomes available. The evaluation uses four origins: 1 January 2009, 1 January 2013, 1 March 2019, and 1 June 2025. These dates were selected to represent distinct interest-rate and macroeconomic regimes: the 2008/2009 financial crisis, the subsequent low-interest-rate period, the pre-COVID environment, and the aftermath of the post-2022 tightening cycle, in order to test model robustness across qualitatively different economic conditions. The earliest origin is constrained by the training requirement of $L + H = 200$ observations per sequence discussed in Section 1.1.

1.6 Evaluation Metrics

Performance is evaluated for both the predicted first differences and the reconstructed deposit-volume levels. Let $y_{k,h}$ and $\widehat{y}_{k,h}$ denote the observed and predicted volume at horizon h after origin o_k. The metrics are

$$\mathrm{MAE}_k = \frac{1}{H} \sum_{h=1}^{H} \left| y_{k,h} - \widehat{y}_{k,h} \right|,$$

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{H} \sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2},$$

$$R_k^2 = 1 - \frac{\sum_{h=1}^{H} \left(y_{k,h} - \widehat{y}_{k,h} \right)^2}{\sum_{h=1}^{H} \left(y_{k,h} - \overline{y}_k \right)^2},$$

where $\overline{y}_k$ is the mean observed volume in evaluation window k. Negative R^2 values indicate performance below the test-window mean benchmark.

A scale-free measure. Absolute errors in EUR billions are not comparable across our four origins, because the deposit stock itself grows strongly over the sample: an error of 10 Bn. € in 2009 represents a far larger relative deviation than the same error in 2025. Averaging absolute MAE across origins therefore implicitly overweights the later, higher-level windows, and the resulting standard deviation conflates genuine model instability with a pure level effect. Following the logic of scaled error measures (hyn), we additionally report the error relative to the constant-level benchmark at the same origin,

$$\text{relMAE}_k = \frac{\text{MAE}_k^{\text{model}}}{\text{MAE}_k^{\text{constant-level}}}, \tag{3}$$

so that relMAE = 1 denotes parity with a forecaster who assumes nothing changes, and values below one denote improvement. This quantity is dimensionless, comparable across regimes, and is the measure on which we base our answer to RQ1.

Reporting. Metrics are calculated separately at each forecast origin and summarized by their mean and standard deviation across origins; the complete per-origin matrix is reproduced in Appendix A rather than only aggregated, since with four observations an aggregate conceals more than it reveals. The primary comparison uses MAE on the reconstructed levels and relMAE; RMSE and R^2 provide complementary information, and metrics on first differences serve as diagnostics. A Diebold–Mariano test (die) with the small-sample correction of **(author?)** (har) was implemented but is deliberately not applied to the main results: with four origins the test has essentially no power, and reporting a non-rejection would invite the false inference that the models are equivalent. Model comparisons are therefore reported descriptively throughout.

2 Results

2.1 Forecasting Performance

Table 3 reports the forecasting performance of each model. The covariate specification with the lowest mean level MAE across the four origins, together with the scale-free measure of Equation (3) is specified. The selection of the covariate set is discussed critically in Section 2.5.

Table 3: Forecasting performance across the four rolling origins (2009, 2013, 2019, 2025). With MAE and RMSE measured in EUR billions and reported as mean ± standard deviation across origins.

Model	Best covariate set	MAE	RMSE	Mean R^2	relMAE
Estimated models					
Ridge	All covariates	9.36 ± 11.11	10.26 ± 12.08	0.178	0.44 ± 0.31
TFT	Macroeconomic	10.78 ± 12.61	12.71 ± 14.18	−0.361	0.49 ± 0.39
FFN	Macroeconomic	11.85 ± 6.90	15.22 ± 10.35	−0.605	0.71 ± 0.34
XGBoost	Excluding inflation	12.71 ± 13.28	14.90 ± 14.79	−0.784	0.69 ± 0.64
Naive benchmarks					
One-step drift	None	11.78 ± 12.81	14.41 ± 14.25	−0.564	0.60 ± 0.43
Constant level	None	17.38 ± 10.13	21.03 ± 10.52	−2.246	1.00

Note: MAE and RMSE are calculated on reconstructed deposit-volume levels. relMAE is the MAE relative to the constant-level benchmark at the same origin; values below one indicate improvement over that benchmark.

Ridge achieved the lowest average MAE and was the only approach with a positive mean R^2. The TFT ranked second, followed by the one-step drift benchmark. Neither the FFN nor XGBoost improved on that simple benchmark on average, and the constant-level forecast produced the largest errors. Greater model complexity therefore did not consistently translate into higher forecasting accuracy.

Answering RQ1. Forecast accuracy is determined mainly by the relMAE, since absolute errors are affected by the level of the deposit stock. Over the 100-day horizon, Ridge records an error equal to 44% of the constant-level benchmark. The one-step drift benchmark records 60%, while XGBoost and the FFN record 69% and 71%, respectively. Thus, all methods improve on the constant-level benchmark, although Ridge's improvement over the one-step drift forecast is smaller. Figure 1 also shows that the standard-deviation bars overlap almost completely, which is the visual form of the statistical caveat stated in Section 1.6: with four origins, this ranking describes a sample, not a population.

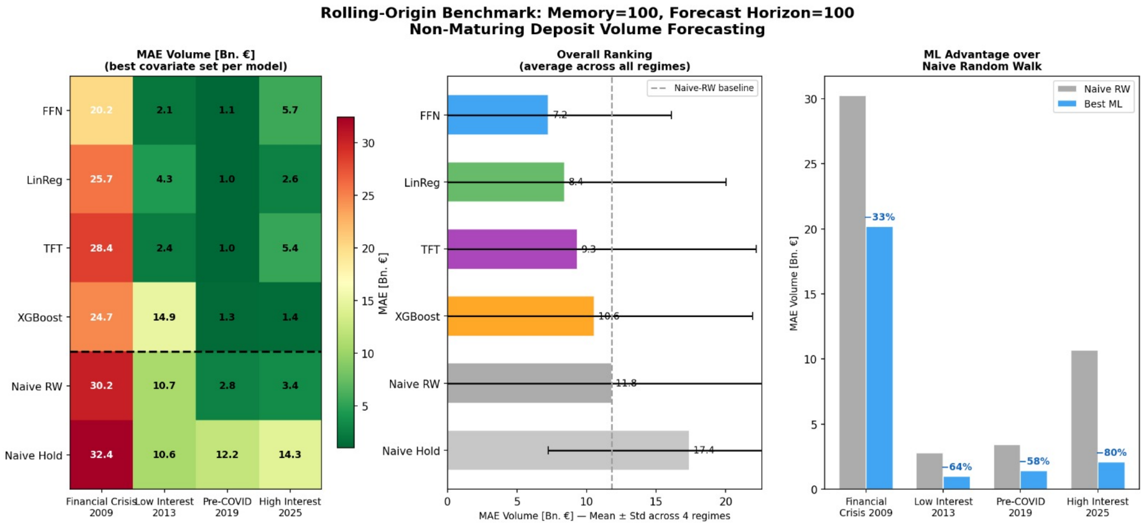

Figure 1: Mean MAE of reconstructed deposit-volume levels across the four forecast origins (2009, 2013, 2019 and 2025). Each model is shown using its lowest-MAE covariate specification, and error bars indicate one standard deviation across origins. The horizontal line marks the one-step drift benchmark.

2.2 Accuracy Versus Robustness

Figure 2 compares forecast accuracy and stability for all model-covariate combinations. Ridge has the most specifications lying on the lower-left region, indicating both lower mean MAE and lower dispersion than the one-step drift benchmark. Ridge is the most accurate model across covariate specifications, whereas the FFN specification using macroeconomic covariates has the lowest absolute dispersion (±6.90).

However, after normalising errors by the constant-level benchmark, Ridge achieves the lowest dispersion among the selected models (0.31 in relMAE), followed by the FFN (0.34) and XGBoost (0.64). Moreover, XGBoost achieves an average relMAE of 0.69, compared with 0.71 for the FFN. This shows why absolute errors are not directly comparable across forecast origins with different deposit levels.

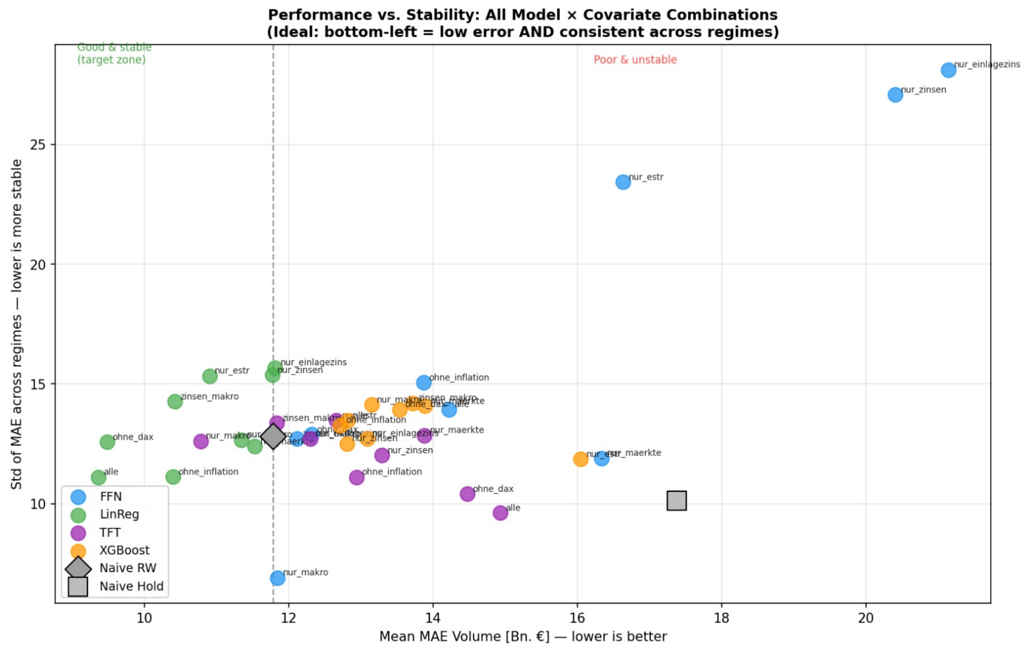

Figure 2: Forecast accuracy and stability across all model–covariate combinations. The horizontal axis shows mean level MAE across the four forecast origins, while the vertical axis shows its standard deviation. Lower values indicate greater accuracy and stability. The dashed lines mark the mean MAE and standard deviation of the one-step drift benchmark.

2.3 Effect of Covariate Selection

To evaluate the effect of the information set, each model is estimated using the nine covariate specifications defined in Table 2. The complete results for all model–covariate combinations are reported in Appendix ??. Figure ?? summarises the lowest-MAE specification for each model.

Ridge performed best with all six explanatory variables (9.36), closely to the specification excluding the DAX (9.48). The FFN and TFT both obtained their lowest MAE using only geopolitical risk and inflation (11.85 and 10.78 respectively). XGBoost performed best when inflation was excluded (12.71).

Results Summary: Rolling-Origin Benchmark
Memory=100 days, Forecast Horizon=100 days, 4 Interest Rate Regimes (2009–2025)

Rank	Model	Best Covariate Set	Mean MAE [Bn. €]	± Std	Mean R²	vs. Naive RW
#1	FFN	nur_makro	7.25	±8.84	0.181	−38%
#2	LinReg	alle	8.41	±11.61	0.255	−29%
#3	TFT	nur_makro	9.33	±12.87	-0.006	−21%
#4	XGBoost	nur_einlagezins	10.56	±11.37	-0.307	−10%
#5	Naive RW	naive	11.78	±12.81	-0.564	+0% worse
#6	Naive Hold	naive	17.38	±10.13	-2.246	+48% worse

Figure 3: Mean level MAE (EUR billions, vertical axis) of each model across the nine covariate specifications of Table 2, averaged over the four rolling origins. The lowest bar per model identifies the specification reported in Table 3.

The FFN achieves the lowest mean MAE using the macroeconomic specification, consisting of geopolitical risk and inflation, with an error of 7.25 Bn. EUR. The linear model performs best when all six explanatory variables are included, reaching a mean MAE of 8.41 Bn. EUR. The TFT also selects the macroeconomic specification and records a mean MAE of 9.33 Bn. EUR, whereas XGBoost performs best using only the deposit rate, with a mean MAE of 10.56 Bn. EUR.

Relative to the one-step drift benchmark, the FFN reduces mean MAE by 38%, the linear model by 29%, the TFT by 21%, and XGBoost by 10%. Therefore, every estimated model improves on the drift benchmark when evaluated using its preferred covariate specification. However, the results also show that no single information set is uniformly optimal: the linear model benefits from the full covariate set, both neural-network models favour the more compact macroeconomic specification, and XGBoost performs best with a single interest-rate variable.

The complete results in Appendix **??** show that forecast accuracy varies across covariate specifications and that this sensitivity is model-dependent. With a lookback window of $L = 100$, every additional covariate contributes 100 further input values before additional lag features are considered. Therefore, expanding the information set substantially increases the input dimensionality of the flexible nonlinear models. By contrast, the ℓ_2 penalty used by the linear model can reduce the dimensionality of this model.

These findings describe how the different model classes exploit the available information in this sample and therefore, should not be interpreted as evidence that the selected variables causally determine household overnight deposit volumes.

2.4 Variation Across Forecast Origins

Forecast accuracy varies considerably across the four evaluation periods. Over the six methods reported in Table 3, the average mean level MAE is approximately 27.9 Bn. EUR at the 2009 origin, compared with 3.6, 8.3, and 9.4 Bn. EUR at the 2013, 2019, and 2025 origins, respectively. Clearly, the 2009 period accounts for a substantial part of the dispersion in the aggregate results. This origin combines the global financial crisis with the smallest available training sample. Given the $L = 100$ lookback and $H = 100$ forecast horizon, the effects of the economic regime and limited training history cannot be separated completely.

Both figures **??**–**??** show that no model dominates across all forecast origins. The FFN records the lowest error in 2009, Ridge performs best in 2013 and remains competitive in 2019 and 2025, while the TFT records the lowest error at the 2025 origin. Moreover, the best-performing covariate specification changes across origins for every model. The aggregate rankings should then be interpreted as average performance across different economic periods, rather than as evidence that one model or covariate specification is consistently superior.

2.5 Interpreting the Selection Procedure

Reported model rankings must be interpreted cautiously because of three features.

First, for each estimated model, the reported value is the lowest mean MAE across nine covariate specifications, selected using the same four rolling origins used for evaluation. The naive benchmarks enjoy no such freedom, so the reported margin of Ridge over the one-step drift benchmark (9.36 versus 11.78, or 20.5%) is an upper bound on the margin one should expect out of sample.

Second, the four origins were deliberately chosen to represent different economic periods rather than sampled randomly. The results therefore cannot be assumed to represent all possible forecast periods.

Third, standard deviations based on only four origins are estimated imprecisely. A formal Diebold-Mariano comparison of forecast accuracy would also have very low power with so few independent origin-level observations.

The results therefore do not establish that Ridge is always superior. However, they provide no evidence that greater model complexity consistently improves accuracy: TFT only slightly

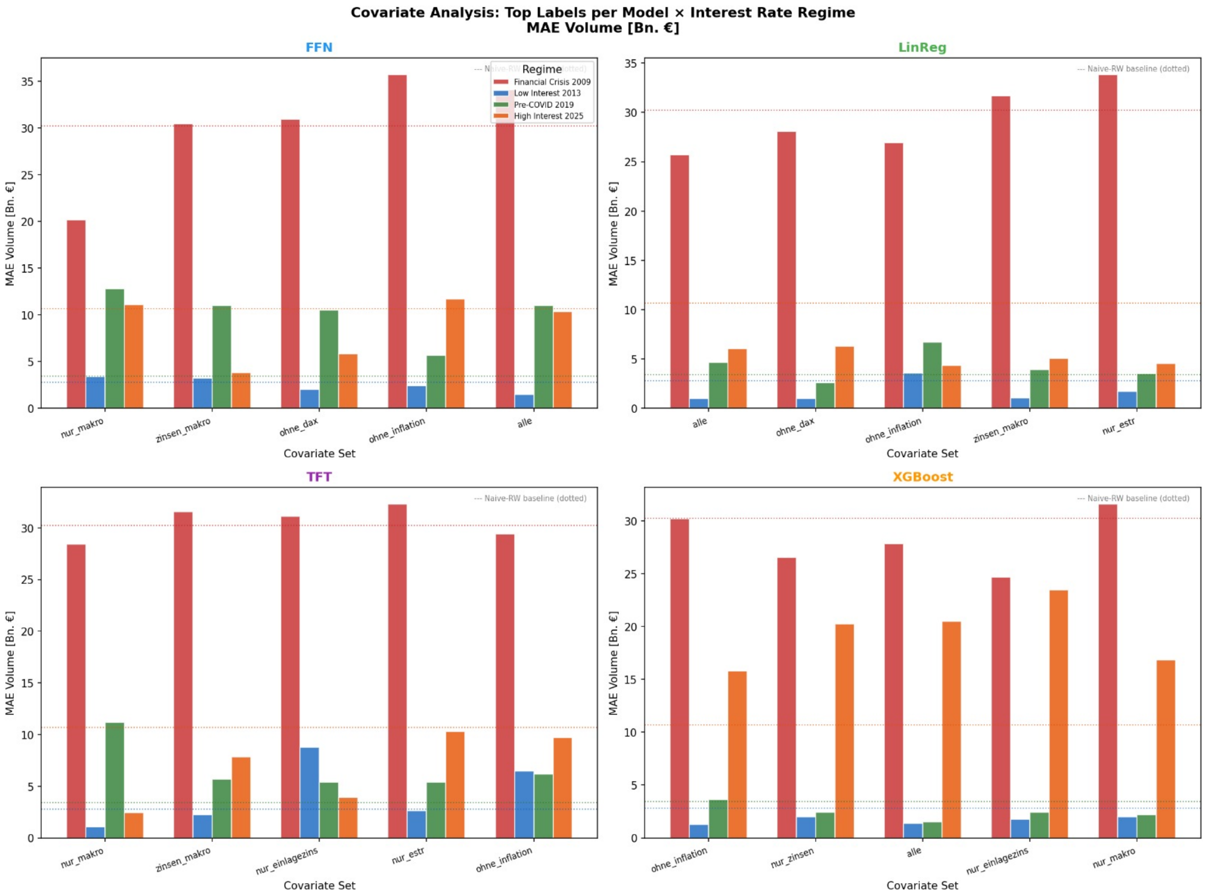

Figure 4: Level MAE (EUR billions) per covariate specification, resolved by forecast origin (2009, 2013, 2019, 2025). Each panel corresponds to one origin; lower values indicate better forecasts. Note the differing vertical scales across panels.

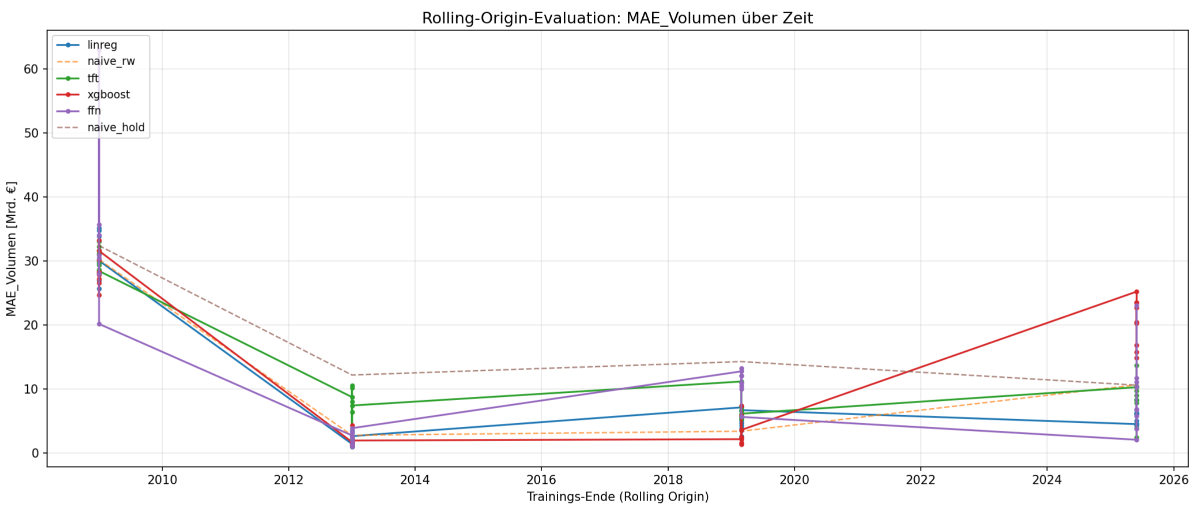

Figure 5: Rolling-origin evaluation: level MAE (EUR billions, vertical axis) per model plotted against the end of the training window (horizontal axis). Naive benchmarks are drawn as dashed reference lines, estimated models as solid lines with markers.

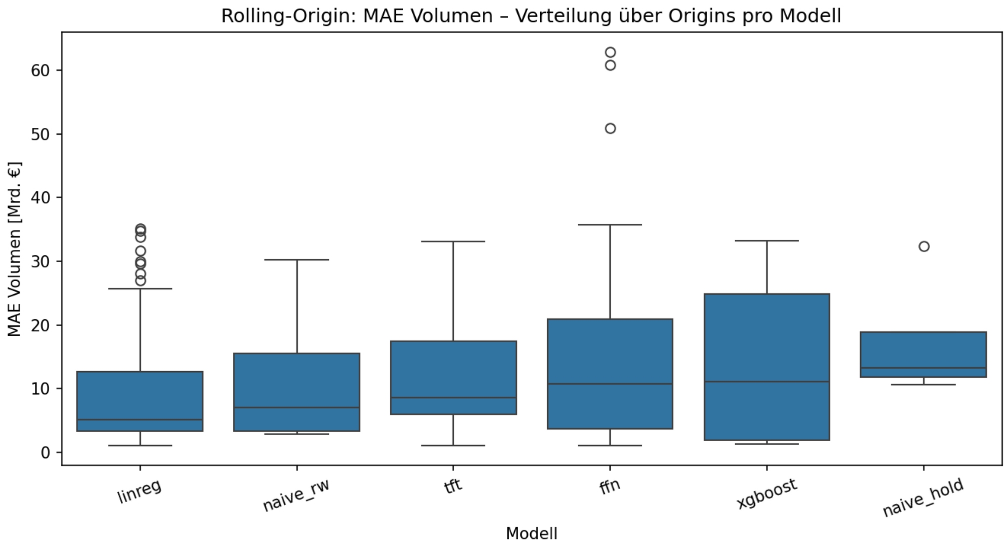

Figure 6: Distribution of level MAE (EUR billions, vertical axis) across the four rolling origins per model, shown as boxplots ordered by median. Box height reflects the interquartile range, whiskers the full observed range across origins.

outperforms the drift benchmark, while the FFN and XGBoost do not, and none of them outperforms Ridge on average.

References

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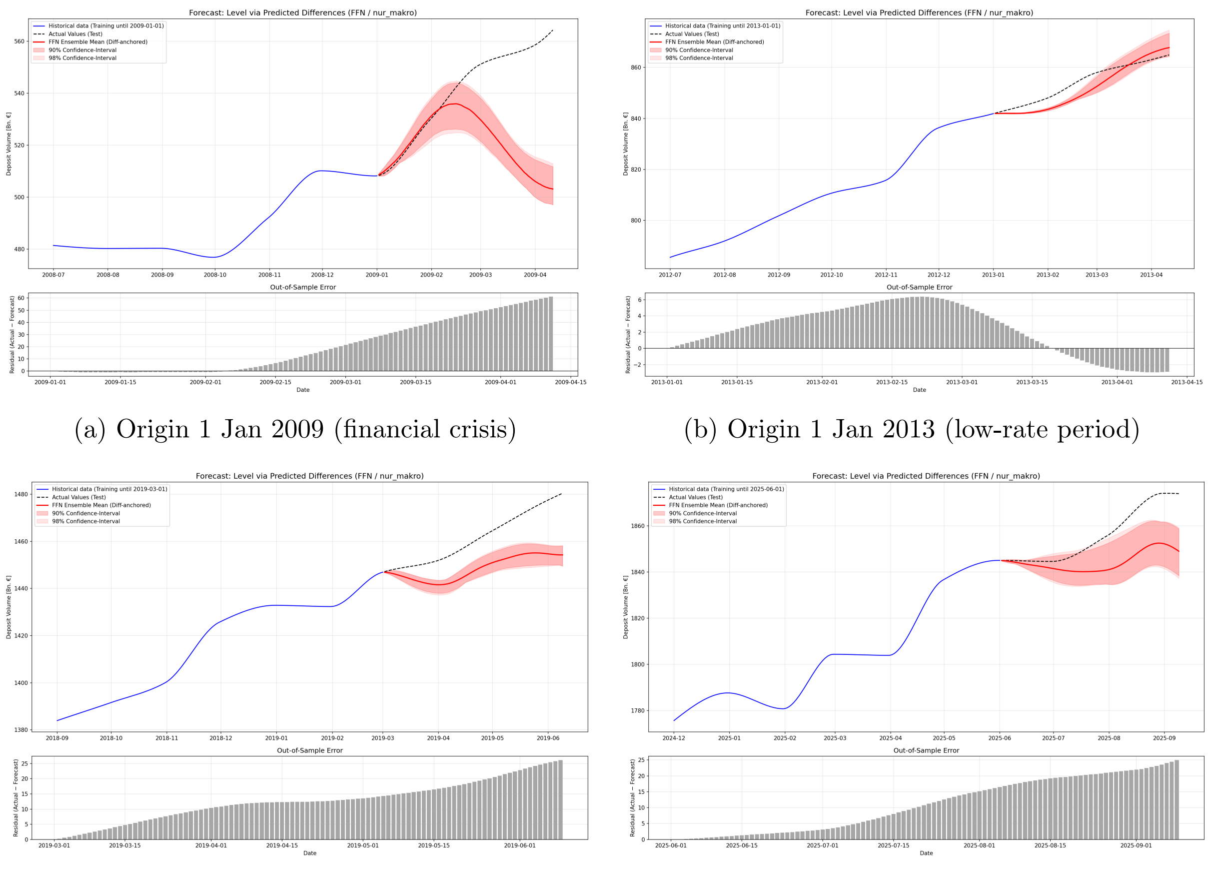
A Appendix: Additional Figures and Tables

B Appendix: Code

The complete implementation, including the rolling-origin driver, the tuning protocol of Table 1, and the scripts producing every figure in this paper, is available in the accompanying repository. The excerpt below shows the interface of the shared model-fitting routine.

Table 4: Complete per-origin results for each model in its best covariate specification. Level MAE in EUR billions; relMAE in parentheses is the ratio to the constant-level benchmark at the same origin. With only four origins, we report the full matrix rather than aggregates alone.

Model	Covariate set	2009-01-01	2013-01-01	2019-03-01	2025-06-01
Ridge	All covariates	25.71 (0.79)	1.00 (0.08)	4.67 (0.33)	6.05 (0.57)
TFT	Macroeconomic	28.44 (0.88)	1.05 (0.09)	11.19 (0.78)	2.43 (0.23)
FFN	Macroeconomic	20.18 (0.62)	3.36 (0.28)	12.77 (0.89)	11.08 (1.04)
XGBoost	Excluding inflation	30.20 (0.93)	1.26 (0.10)	3.63 (0.25)	15.77 (1.49)
One-step drift	None	30.23 (0.93)	2.80 (0.23)	3.43 (0.24)	10.68 (1.01)
Constant level	None	32.40 (1.00)	12.20 (1.00)	14.30 (1.00)	10.62 (1.00)
Mean across models	—	27.86	3.61	8.33	9.44



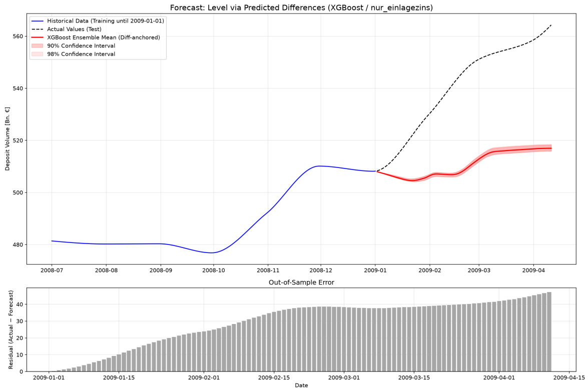
(a) Origin 1 Jan 2009 (financial crisis)

(b) Origin 1 Jan 2013 (low-rate period)

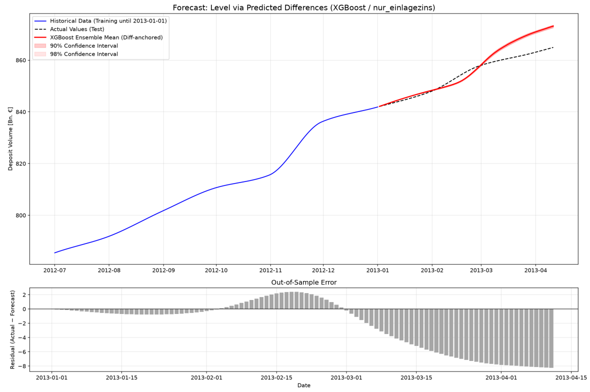
(c) Origin 1 Mar 2019 (pre-COVID)

(d) Origin 1 Jun 2025 (post-tightening)

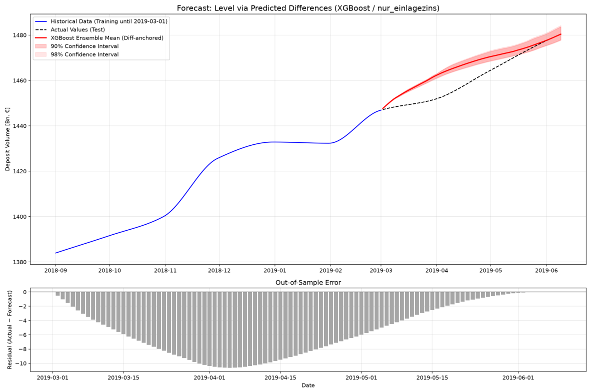
Figure 7: FFN forecasts with the macroeconomic covariate set at all four rolling origins; panel construction as in Figure ??.



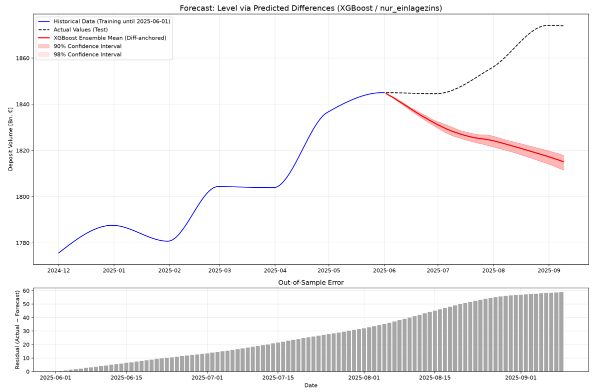
(a) Origin 1 Jan 2009 (financial crisis)



(b) Origin 1 Jan 2013 (low-rate period)



(c) Origin 1 Mar 2019 (pre-COVID)



(d) Origin 1 Jun 2025 (post-tightening)

Figure 8: XGBoost forecasts with the deposit-rate-only specification at all four rolling origins; panel construction as in Figure ??. This is not the best-performing XGBoost specification (see Table 3) but illustrates the extrapolation behaviour discussed in Section ?? most clearly.

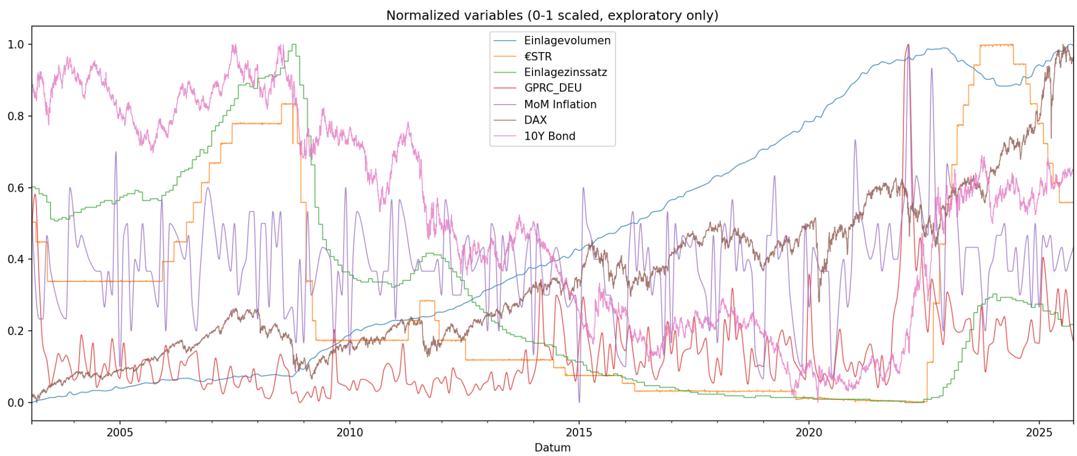


Figure 9: All variables after standardisation to zero mean and unit variance over the sample period January 2003 – September 2025, shown on a common vertical scale so that co-movement between the deposit volume and the covariates is directly visible.

Listing 1: Shared model-fitting interface.

```python
def fit_sequence_model(model_name, X_train, y_train, seed=SEED,
                       bootstrap=False, do_tune=False,
                       tuned_estimator=None):
    """Fit␣one␣ensemble␣member␣of␣a␣sequence␣model.

␣␣␣␣model_name␣␣␣␣␣␣:␣"linreg"␣|␣"ffn"␣|␣"xgboost"
␣␣␣␣bootstrap␣␣␣␣␣␣␣:␣resample␣training␣sequences␣(ensemble␣members␣S␣>␣1)
␣␣␣␣do_tune␣␣␣␣␣␣␣␣␣:␣run␣the␣shared␣grid␣search␣(<=␣TUNING_BUDGET␣configs)
␣␣␣␣tuned_estimator␣:␣reuse␣a␣cached␣configuration␣for␣the␣same␣origin
␣␣␣␣"""
    ...
```